BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: 9231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 9231

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. If A and B are two polynomials and their HCF and LCM are respectively H and L then A is equal to:

(A) H×L/B
(B) B/H×L
(C) B/L
(D) B/H

Q2. If A={1,2} and B={3,4} then which of the following is ordered pair of A×B?

(A) {(3,2)}
(B) {(1,2)}
(C) {(2,3)}
(D) {(4,2)}

Q3. The solution of linear equation 2x + 3 = 1 − 7x is:

(A) −4/5
(B) −2/9
(C) 4/5
(D) 2/9

Q4. Which of the following elements represent one of the rows of the matrix [1 2 3; 6 −1 3; 2 4 9]?

(A) 1, 6, 2
(B) 6, −1, 3
(C) 1, −1, 9
(D) 4, 3, 9

Q5. Which of the following is the base of natural logarithm?

(A) 2
(B) π
(C) e
(D) 10

Q6. Which of the following is a rational expression?

(A) $(\sqrt{(5x^2+3x-2)})/(10-x)$ (where $10-x\neq0$)

(B) $(\sqrt{(8x^3+8x+2)})/(x+5)$ (where $x+5\neq0$)

(C) $(\sqrt{(7x^2+4x+7)})/(x+7)$ (where $x+7\neq0$)

(D) $(\sqrt{(3x^2+2x-5)})/(7-x)$ (where $7-x\neq0$)

Q7. $(2x+y)^3$ is equal to:

(A) $8x^3+y^3+6xy(2x+y)$

(B) $8x^3-y^3-6xy(2x-y)$

(C) $8x^3+y^3-6xy(2x-y)$

(D) $8x^3-y^3-6xy(2x+y)$

Q8. The factorization of $x^2+xy+8x+8y$ is:

(A) $(x+y)(x+8)$

(B) $(2x+y)(x+8)$

(C) $(x+2y)(x+8)$

(D) $(x+y)(2x+8)$

Q9. Which of these two matrices are equal? F=[1 3; 4 2] G=[7−3 3; 6 2] H=[4−3 3; 2+2 2] J=[4−3 6; 3+3 7]

(A) F and H

(B) F and G

(C) G and H

(D) G and J

Q10. If $z=5+7i$ then conjugate of z is:

(A) $-5-7i$

(B) $-5+7i$

(C) $5-7i$

(D) $-7i$

Q11. Which of the following is a row matrix?

(A) [8 9 −4 5]

(B) [8; 9; −4; 5]

(C) [8 9; −4 5]

(D) [8; 9; −5]

Q12. $\sqrt[8]{y^3}$ can be expressed in exponential form as:

(A) y^3

(B) $8y^3$

(C) $y^{8/3}$

(D) $y^{3/8}$

Q13. 0.000327 can be written in scientific notation as:

(A) 3.27×10^{-4}

(B) 3.27×10^{4}

(C) 0.0327×10^{-4}

(D) 32.7×10^{2}

Q14. If P=[−7 5; 4 −3] and Q=[2 −3; 3 4] then P+Q is equal to:

(A) [9 8; 7 7]

(B) [−5 2; 7 1]

(C) [5 −2; 7 −1]

(D) [−9 −2; 7 −7]

Q15. The irrational number √11 can be expressed as:

(A) 3.3166...

(B) 33.166...

(C) 331.66

(D) 0.3316

SECTION-B

Time Allowed: 2 Hours 40 Minutes	**Maximum Marks:** 36	**Paper Code:** 9231

Q1. Attempt any nine of the following short questions. Each question carries 4 marks.

(i) If A=[6 2; 3 4] and B=[5 −1; −2 3] then determine whether AB=BA or not.

(ii) Simplify: (i) $(-3a^3b^2)^2$ (ii) $(-2xy^2)^3$

(iii) Find the value of 'a': $\log_2 6+\log_2 7=\log_2 a$

(iv) Evaluate the following when a=−2: $(2a+1)/(4a^2-1)$

(v) if a+b=10, a−b=10 then find the value of a^2+b^2

(vi) Without performing division, find the remainder when $2t^3-3t^2+t-2$ is divided by t−2

(vii) Simplify: (a+b)/(3a+2b) + (a−b)/(3a+2b)

(viii) Solve: $20-3\sqrt{t} = \sqrt{t}+4$

(ix) Solve and plot on number line: 3(t−2)>15

(x) Prove that the points A(6,−2), B(2,2) and C(−2,6) are collinear.

(xi) Prove that: If two angles of a triangle are congruent, then the sides opposite to those angles are also congruent.

(xii) Prove that: Any point on the right bisector of a line segment is equidistant from end points of the segment.

SECTION-C

Maximum Marks: 24

> Note: Attempt any four of the following questions. Each question carries 6 marks.

Q2. Prove that: If two opposite sides of a quadrilateral are congruent then it is a parallelogram.

Q3. Prove that: From a point outside a line, the perpendicular is the shortest distance from the point to the line.

Q4. Prove that: If a line segment intersects the two sides of a triangle in the same ratio then it is parallel to the third side.

Q5. Prove that: In a right angled triangle, the square of the length of hypotenuse is equal to the sum of the squares of the lengths of the other two sides.

Q6. Prove that: Parallelograms on equal bases and having the same altitude are equal in area.

Q7. Construct ΔABC such that $mBC=3.3cm$, $mAB=6.2cm$ and $m\angle A=30°$